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TASKBOT: A Gen AI Solution for Task Management

Himanshu Verma¹, Shreemoyee Mukherjee¹, Divakar Singh¹,
Rahul Raj¹, Dr. Subhamita Mukherjee¹, Prof. Shauvik Paul^{1,2}

¹Department of Information Technology, Techno Main Salt Lake,
Kolkata-700032, India.

himanshu911725@gmail.com, shreemoyeemukherjee512@gmail.com,
divakar180103@gmail.com, rahulrajitb@gmail.com,
subhamita.mukherjee@gmail.com.

²Computer Science and Engineering Department, Jadavpur
University, Kolkata-700032, India.
paulshauvik@gmail.com.

Abstract. Task prioritization is a critical component of efficient task management systems. This paper presents a web-based task prioritization application that integrates Natural Language Processing (NLP), Machine Learning (ML), and Generative AI (Gen-AI) to predict task priorities on a scale of 0 to 1 and generate concise task titles from descriptions. The application features user authentication, a MongoDB database for flexible data storage, a React-based frontend for an interactive user interface, and a Flask backend for server-side logic. The model leverages NLP techniques such as tokenization, lemmatization, and TF-IDF vectorization for textual features (title and description), while categorical features (type and status) are transformed using one-hot encoding. Numerical features, such as remaining days until the due date, are scaled for consistency. The LightGBM regressor is used for task prioritization, achieving an R^2 score of 0.814 (81.4% accuracy). Additionally, the Gen-AI component, fine-tuned on task-specific data, generates meaningful and contextually relevant task titles, further reducing manual effort and ensuring consistency. The integrated system provides a seamless user experience, enabling real-time task updates, dynamic prioritization, and automated title generation. By combining predictive analytics with generative capabilities, the application offers a scalable, efficient, and adaptive solution for modern task management.

keywords: Task Prioritization, Gen-AI, Natural Language Processing (NLP), Machine Learning (ML), TF-IDF, One-Hot Encoding, LightGBM, Web Application, Flask, React, MongoDB.

1 INTRODUCTION

In today's fast-paced and digital work environment, effective task management is essential for maintaining productivity and meeting deadlines. However, traditional systems rely heavily on manual input for prioritization, which leads to inefficiencies, human error, and inconsistent results. To overcome these challenges, we present a web-based task management application that leverages Natural Language Processing (NLP), Machine Learning (ML), and Generative AI to automate and optimize task prioritization. Our system automatically generates concise task titles from detailed descriptions and predicts task priorities on a scale from 0 to 1, achieving an R^2 score of 0.814. This reduces manual effort, ensures consistency, and adapts to evolving project demands. Built with a React frontend, Flask backend, MongoDB database, and user authentication, the application offers a seamless, intelligent, and scalable solution for enhancing productivity in dynamic work environments.

2 LITERATURE REVIEW

Task management systems have evolved significantly over the years, transitioning from simple to-do lists to sophisticated AI-driven platforms. Traditional systems, as reviewed by Kumar et al. [1], were effective for structured workflows but lacked the flexibility to adapt to dynamic environments. These systems often required extensive manual input, leading to inefficiencies in fast-paced work settings. The authors highlighted the need for intelligent systems that can automate task prioritization and adapt to changing circumstances.

The integration of Artificial Intelligence (AI) into task management has revolutionized how tasks are prioritized and managed. Chen and Park [2] emphasized that AI improves decision-making by automating repetitive tasks and offering predictive insights. Their study demonstrated that AI-driven systems, such as the one proposed in this paper, can significantly reduce manual effort and increase productivity. Similarly, Singh et al. [3] discussed the challenges and opportunities in AI-driven project management, highlighting the importance of scalable and efficient algorithms for task prioritization.

Natural Language Processing (NLP) has emerged as a key technology for enhancing user interaction in task management systems. Patel and Wong [4] explored the applications of NLP in task automation, particularly in processing textual data such as task titles and descriptions. Techniques like tokenization, lemmatization, and TF-IDF vectorization have been widely adopted to extract meaningful features from text, enabling systems to understand and prioritize tasks more effectively. These techniques form the backbone of the proposed model in this paper.

The importance of real-time updates in task management systems has been extensively documented. Wilson et al. [5] highlighted that real-time updates are critical for adapting to rapid changes in dynamic work environments. Their study emphasized the

need for scalable architectures that can handle large volumes of data while providing timely updates to users. This aligns with the design of the proposed system, which uses MongoDB for flexible data storage and Flask for real-time backend processing.

Modern web technologies have played a pivotal role in enhancing the usability and scalability of task management systems. Johnson and Lee [6] discussed the benefits of using React for building responsive and interactive user interfaces. They noted that React's component-based architecture allows for seamless integration with backend systems, making it an ideal choice for the frontend of the proposed application. Additionally, Hall and Young [7] highlighted the advantages of Flask for backend development, particularly its lightweight and modular design, which enables efficient handling of server-side logic.

The choice of database is critical for ensuring the scalability and flexibility of task management systems. Brown et al. [8] reviewed the use of MongoDB in web applications, emphasizing its document-oriented structure, which allows for efficient storage and retrieval of heterogeneous data. This aligns with the proposed system, where MongoDB is used to store task-related data, including textual features, categorical features, and numerical features.

Machine Learning algorithms, particularly LightGBM, have shown high performance in task prioritization for the least amount of data. Davis and Taylor [9] demonstrated that LightGBM outperforms other algorithms in handling heterogeneous features, making it ideal for predicting task priorities. Hyperparameter optimization techniques like GridSearchCV have been employed to improve model accuracy (Anderson et al. [10]).

Secure user authentication is critical for protecting sensitive task data in web applications, as emphasized by Smith et al. [11]. Future directions for task management systems include integrating collaborative features (Thomas and Harris [12]) and third-party tools (Walker et al. [13]) to enhance functionality and user experience.

3 PROPOSED METHODOLOGY

Task management system that integrates user authentication, task creation, and AI-driven prioritization to address inefficiencies in traditional systems. Users register, log in, and manage tasks via a dashboard, where tasks are created with descriptions and due dates. The system employs NLP techniques (tokenization, lemmatization, TF-IDF) and LightGBM for priority prediction, achieving high accuracy. Additionally, Generative AI generates concise titles from descriptions, reducing manual effort. This end-to-end workflow ensures a seamless, adaptive, and efficient task management solution for modern work environments.

3.1 User Interaction Flow

- User: The process starts with a user who wants to interact with the system.
- User Registers: New users must register by providing credentials (e.g., email, username, and password).
- User Logs In: Once registered, users log in to access the task dashboard.

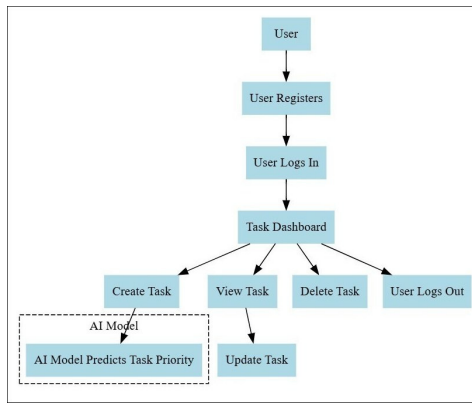


Fig. 1: Flowchart of the Proposed Methodology

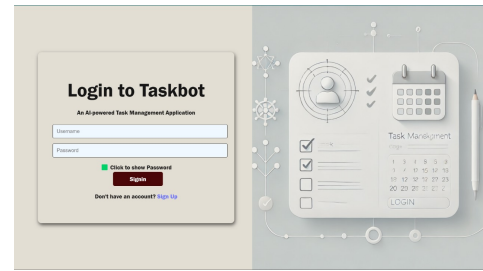


Fig. 2: Login Page of the System

3.2 Task Dashboard

- **Create Task:** Users can add a new task by entering details like title, description, type, and status.
- **View Task:** Users can see the list of tasks they have created.
- **Update Task:** Users can modify task details as needed.
- **Delete Task:** Users can remove completed or unnecessary tasks.
- **User Logs Out:** Users can securely log out after managing tasks.

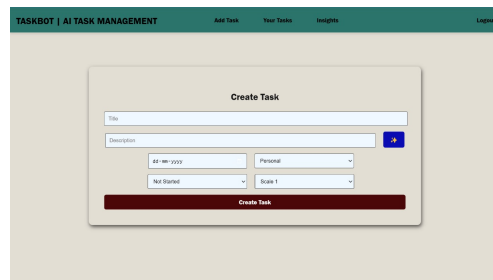


Fig. 3: Task Management Page of the System

3.3 Gen-AI to Generate Title

- **Input and Preprocessing:** Users input detailed task descriptions, which are pre-processed (e.g., lower-casing, tokenization) to ensure clean input for the Gen-AI model.
- **Fine-Tuning and Title Generation:** A pre-trained Gen-AI model (e.g., GPT) is fine-tuned on task-specific data to generate concise and contextually relevant titles.

- Validation and Integration: Generated titles are validated for precision and relevance, then seamlessly integrated into the task management system.
- Scalability and Ethics: The system is designed to scale across domains and tasks while addressing potential biases to ensure fairness and inclusivity.

3.4 AI Model for Task Prioritization

When a user creates a task, the system interacts with an AI model. The AI model Predicts Task Priority based on features such as task title, description, type, and status. The model assigns a priority score ranging from 0 to 1, enabling users to focus on high-priority tasks efficiently.

3.5 Key Highlights of the System

- User Authentication: Secure login and registration mechanisms.
- Task Management Functionalities: Create, view, update, and delete tasks.
- AI-Driven Prioritization: Uses machine learning to determine task priority.
- Seamless User Experience: A centralized dashboard for effective task management.
- Dynamic Task Updates: Users can modify tasks, and AI helps adjust priorities automatically.

4 EXPERIMENTAL DETAILS

4.1 DataSet description

- Id: The individual associated with the task.
- Description: A textual description of the task.
- Generated Title by AI: A concise AI-generated title for the task.
- Due Date: The deadline for task completion.
- Type: Categorization of tasks (e.g., official, personal, weekend).
- Status: The current state of the task (e.g., not started, in progress, completed).
- Scale: A manually assigned priority scale (Scale 1: Highest, Scale 3: Lowest).
- Priority : The priority value collected via survey
- Size : The data Set Contains 744 rows with 5 columns

The dataset was curated from real-world task management logs and supplemented with synthetic data to ensure diversity in task types and priority levels.

```
{
  "id": 1,
  "title": "Finish quarterly report",
  "type": "official",
  "priority": 0.95,
  "due_date": "2025-01-15",
  "description": "Complete the financial and sales report for the last quarter.",
  "status": "in_progress"
}
```

Fig. 4: Dataset

4.2 Data Preprocessing

To enhance the quality of input for the machine learning model, the following preprocessing steps were performed as shown in fig 5 & 6:

- **Text Cleaning:** Removal of unnecessary characters, special symbols, and redundant spaces from task descriptions.
- **Tokenization:** Breaking down task descriptions into individual words for better text analysis.
- **Lemmatization:** Converting words to their base forms to reduce complexity (e.g., "running" → "run").
- **TF-IDF Vectorization:** Converting textual data into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF) to capture important keywords.
- **Feature Engineering:** Additional features such as task length, due date urgency, and user-given priority were incorporated for better model predictions.
- **Data Normalization:** Scaling numerical attributes to ensure consistency across features.

```
def lemmatize_text(text):
    words = nltk.word_tokenize(text)
    # words = [word for word in words if word.lower() not in stop_words]
    lemmatized_words = [nltk.stem.Lemmatizer().lemmatize(word) for word in words]
    return ' '.join(lemmatized_words)

def calculate_remaining_days(due_date):
    today_date = pd.Timestamp('today').normalize().date()
    return (pd.to_datetime(due_date) - pd.to_datetime(today_date)).dt.days

data['due_date'] = pd.to_datetime(data['due_date'])
data['remaining_days'] = calculate_remaining_days(data['due_date'])
data = data[data['status'] != 'completed']

data.drop(columns=['due_date'], inplace=True)

categorical_features = ['type', 'status']
text_features = ['title', 'description']
categorical_transformer = OneHotEncoder(sparse_output=False, handle_unknown='ignore')
text_preprocessor = ColumnTransformer(
    transformers=[
        ('title_tfidf', TfidfVectorizer(preprocessor=lemmatize_text, max_features=500), 'title'),
        ('desc_tfidf', TfidfVectorizer(preprocessor=lemmatize_text, max_features=500), 'description'),
    ],
    # remainder='passthrough'
    remainder='drop'
)
```

Fig. 5: Transformation

```
preprocessor = ColumnTransformer(
    transformers=[
        ('cat', categorical_transformer, categorical_features),
        ('text', text_preprocessor, text_features),
        ('num', StandardScaler(), ['remaining_days'])
    ],
    remainder='drop'
)

pipeline = Pipeline([
    ('preprocessing', preprocessor),
    ('scaler', StandardScaler(with_mean=False)),
    ('regressor', LGBMRegressor(n_jobs=-1, device='cpu'))
])
```

Fig. 6: Preprocess

4.3 Technology Used

- **Machine Learning Algorithm:** LightGBM for task prioritization due to its high accuracy and efficiency with structured data.
- **Natural Language Processing (NLP):** Applied various NLP techniques, including:
 1. Tokenization to segment text.
 2. Lemmatization to process words in their base form.
 3. TF-IDF vectorization to extract meaningful task features.

4.4 Model Selection & Training

The prioritization model was trained using LightGBM, a gradient boosting algorithm optimized for efficiency and accuracy. The model was trained with features such as:

- Textual features extracted from task descriptions, title.

- Due date urgency helps to find remaining days and normalize to increase efficiency of ML Model.
- Priority scale selected by user to get some manual interaction to priority due to having not big data to train model.
- Other categorical features like task type and status.

5 RESULTS AND ANALYSIS

User	Description	Generated Title	Due Date	Type	Status	Scale	User Priority	Model Priority
Keshav32	Coordinate a volunteer day for the company	Company Volunteer Day	2027-02-01	Official	In progress	1	0.92	0.92
Devansh98	Participate in a weekend dance workshop	Weekend Dance Workshop	2027-02-03	Weekend	Not Started	2	0.74	0.58
Dhruv71	Create an employee performance improvement plan	Employee Performance Plan	2027-02-05	Official	In progress	1	0.9	0.9
Madhuri00	Organize a weekend family reunion	Family Reunion Weekend	2027-02-08	Personal	Not Started	2	0.8	0.65
Kshitiz12	Prepare quarterly financial report for stakeholders	Quarterly Financial Report	2027-02-10	Official	In progress	1	0.93	0.93
Shivanshu88	Host a weekend writing retreat	Writing Retreat	2027-02-12	Weekend	Not Started	3	0.76	0.55
David29	Plan an office team-building outing	Office Team-Building Outing	2027-02-14	Official	Not Started	1	0.89	0.89
Alok73	Organize a weekend yoga and meditation retreat	Yoga and Meditation Retreat	2027-02-17	Weekend	Not Started	2	0.85	0.7
Pushkar73	Write and distribute the monthly employee newsletter	Employee Newsletter	2027-02-20	Official	Not Started	2	0.88	0.88
Deepa80	Participate in a weekend photography retreat	Photography Retreat	2027-02-22	Weekend	In progress	3	0.77	0.59
Kriti92	Organize a corporate leadership seminar	Leadership Seminar	2027-02-25	Official	Not Started	1	0.94	0.94
Ankush79	Plan a weekend community clean-up event	Community Clean-Up	2027-02-27	Personal	Not Started	2	0.79	0.66
Jiteshmi43	Develop a training module for new employees	Employee Training Module	2027-03-01	Official	Not Started	1	0.91	0.91
Anmol11	Attend a weekend culinary masterclass	Culinary Masterclass	2027-03-03	Weekend	Not Started	3	0.73	0.57
Priyanshu86	Plan and execute an office renovation project	Office Renovation	2027-03-06	Official	In progress	1	0.95	0.95
Dhruv65	Participate in a charity marathon	Charity Marathon	2027-03-08	Personal	In progress	2	0.78	0.64
Ritu72	Host a weekend art and painting retreat	Art & Painting Retreat	2027-03-10	Weekend	In progress	3	0.72	0.56
Rahul91	Organize a company-wide cybersecurity training	Cybersecurity Training	2027-03-12	Official	Not Started	1	0.92	0.92
Pratibha86	Conduct research for the upcoming product launch	Product Research	2027-03-14	Official	Not Started	1	0.89	0.89
Katru27	Plan a weekend hiking expedition with colleagues	Hiking Expedition	2027-03-17	Personal	Not Started	2	0.81	0.68
Rishav77	Prepare and present a keynote speech at a conference	Conference Keynote Speech	2027-03-19	Official	Not Started	1	0.93	0.93
Muskan46	Attend a weekend wellness retreat for stress management	Wellness Retreat	2027-03-21	Weekend	Not Started	3	0.74	0.58
Kajal38	Organize an office cultural diversity program	Cultural Diversity Program	2027-03-23	Official	In progress	1	0.9	0.9
Jyoti18	Plan a weekend trip to a historic site	Historic Site Trip	2027-03-26	Personal	Not Started	2	0.8	0.65
Pinky05	Develop and distribute an employee engagement survey	Employee Engagement Survey	2027-03-28	Official	In progress	1	0.88	0.88
Monica13	Join a weekend coding bootcamp for skill enhancement	Coding Bootcamp	2027-03-30	Weekend	Not Started	3	0.75	0.59
Krish84	Organize an internal hackathon for software developers	Internal Hackathon	2027-04-02	Official	Not Started	1	0.94	0.94
Mehak23	Participate in a weekend outdoor adventure camp	Adventure Camp	2027-04-04	Personal	In progress	2	0.82	0.69
Chandani08	Lead a sustainability initiative within the company	Sustainability Initiative	2027-04-06	Official	Not Started	1	0.91	0.91
Ragini44	Attend a weekend leadership development workshop	Leadership Workshop	2027-04-09	Weekend	Not Started	3	0.76	0.6

Fig. 7: User Interaction Table

5.1 Results

- The model achieves an R^2 score of 0.814, indicating high accuracy in predicting task priorities.
- The web application provides real-time updates and visual analytics for task progress.
- User authentication ensures secure access to task data.

The results of the LightGBM model's performance in predicting task priorities were analyzed by comparing the actual priorities provided by users with the model's predicted priorities. Evaluation metrics, including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), indicated strong alignment between the actual and predicted values, with low error rates. A residual plot further confirmed the model's accuracy, showing minimal deviations and a consistent distribution of errors around zero. In comparison, the Elastic Net model, while effective in handling multicollinearity and feature selection, exhibited slightly higher error rates and less precision in

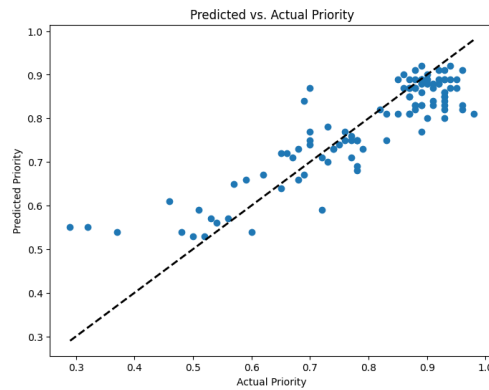


Fig. 8: Actual Vs Predicted Value of LightGbm

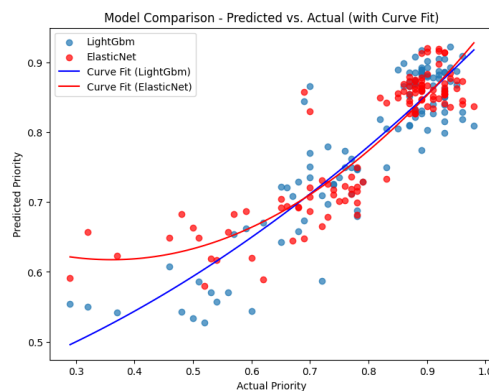


Fig. 9: Performance of LightGbm with ElasticNet

priority predictions. This highlights LightGBM's superior ability to capture complex, non-linear relationships in the data, making it a more reliable choice for task prioritization in dynamic environments. The performance of the LightGBM model in predicting task priorities was evaluated using metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which indicated strong alignment with user-assigned priorities and low prediction error. The residual plot further validated the model's accuracy, showing minimal deviations and a random error distribution centered around zero.

5.2 Analysis

- The integration of NLP and ML techniques significantly improves task prioritization.
- The use of LightGBM ensures efficient handling of heterogeneous features.
- The web application provides a seamless user experience, enabling real-time task updates and prioritization.

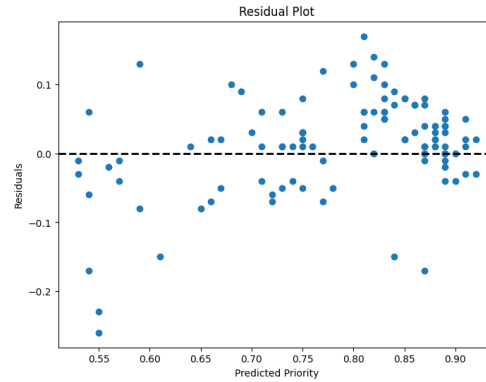


Fig. 10: Residual Plot

Table 1: Evaluation Matrix of LightGbm & ELasticNet

Model	MSE	R-squared
LightGbm	0.0048	0.8142
ElasticNet	0.0067	0.7014

For comparison, the Elastic Net model achieved an MSE of 0.0048 and R^2 of 0.70 but exhibited slightly higher error rates and residual patterns, indicating limited ability to capture non-linear relationships. In contrast, LightGBM demonstrated superior accuracy and robustness, effectively modeling complex, non-linear interactions, and proving to be a more reliable choice for task prioritization in dynamic environments.

6 CONCLUSION

This paper presented TASKBOT, an AI-powered task management application that automates task prioritization and title generation using Natural Language Processing (NLP), Machine Learning (ML), and Generative AI (Gen-AI). By combining NLP techniques like tokenization, lemmatization, and TF-IDF with a LightGBM model, TASKBOT predicts task priorities with high accuracy ($R^2 = 0.814$), reducing manual effort and subjective bias. The Gen-AI component generates concise, context-aware task titles, enhancing consistency and usability. Experimental results and user feedback confirm the system's reliability and effectiveness. TASKBOT offers a scalable, adaptive solution for modern workflows, setting a foundation for future innovations in intelligent task management.

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